Chapter 6 exercises

DS06-E01 • Search order and representation cost

An undirected graph has edges AB, AC, BD, CD, DE, with neighbors in alphabetical order. Run complete BFS and recursive DFS from A. Give visit order, parents, and BFS distances. How does matrix representation change the time bound?

DS06-E02 • Two MST algorithms

Use edges AB=2, AC=5, BC=1, BD=4, CD=3, CE=7, DE=2. Run Kruskal with alphabetical endpoint order for ties, then Prim from A. List edge selections and reasons for rejecting edges. Prove one selection safe using a cut exchange.

DS06-E03 • Negative edges versus negative cycles

Directed edges are s→a=2, s→b=5, b→a=−4, a→t=3. Compare settled-vertex Dijkstra with Bellman–Ford. Then add a→b=1. Identify vertices without finite shortest distances. Are all vertices affected?

DS06-E04 • Floyd–Warshall loop order

Explain the intermediate-vertex invariant with k outermost. Why can the same proof not justify arbitrary loop exchange followed by just one triple-loop pass? Trace a directed path requiring at least two intermediate vertices.

DS06-E05 • AOE critical path

Use activities s→a:3, s→b:2, a→t:4, b→t:6. Compute every E, L, activity total slack, and critical path. If both s→a and a→t grow by 1, does total completion time stay unchanged? Explain.

DS06-E06 • Single-source paths in a DAG with negative edges

Design topological-order relaxation, prove it by induction, and derive O(V+E) time. Identify the failing step if the graph contains a cycle. Negative edges alone do not rule out linear time.

DS06-E07 • Three objectives on one graph and negative initialization

Use undirected edges s–p6,p–t6,s–q2,q–t9,p–q7. Find the shortest-sum path, minimax path, and one MST. Trace both label-setting sequences and parent changes. Prove the minimax property of every MST path. Why is source label 0 wrong for negative bottleneck weights?

DS06-E08 • Mandatory edges and alternatives across all optima

Vertices A,B,C,D form K4 with all six weights 2. Vertices E,F,G form a triangle with all three weights 5. The sole connector D–E has weight 1. Use weight layers and quotient graphs to prove the edge counts at each weight and classify mandatory edges. Construct two MSTs whose symmetric-difference weight exceeds one MST' s weight. Must that difference decompose into vertex-disjoint paths and cycles?

DS06-E09 • Articulation points, bridges, and equality

An undirected graph has edges AB, AC, BC, BD, DE, DF, EF, DG. Start DFS at A with alphabetical neighbor order. List tin and low, all articulation points and bridges. Explain why DE is not a bridge at equality even though D is a cut vertex.

DS06-E10 • DFS root and parallel edges

An undirected multigraph has two distinct A–B edges e_1,e_2 and one B–C edge e_3. Start DFS from A along e_1. Find cut vertices and bridges. If all edges to the parent vertex are skipped, which edge is misclassified? How many DFS children has root B when DFS starts at B?

DS06-E11 • Synchronous relaxation and a reachable negative cycle

Directed edges are s→a=2, s→b=5, b→a=−4, a→t=3. Give synchronous Bellman–Ford distances (s,a,b,t) after rounds 0 through 3. Add a→b=1. Which vertices have distance −∞, and are vertices unreachable from the negative cycle affected?

References and attribution

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https://github.com/xinhi6716/computer-science-self-study